

ENVIRONMENTAL STUDIES - FINAL REVISION NOTES

ENVIRONMENTAL STUDIES – FINAL EXAM NOTES

1. WATER POLLUTION

Definition: Undesirable change in physical, chemical or biological properties of water making it unfit for use.

Sources: Sewage, industrial waste, agricultural runoff, oil spills.

Key Concepts: Eutrophication, Biomagnification.

Control: STP, ETP, afforestation, reduced pesticide use.

2. AIR POLLUTION

Definition: Presence of harmful gases or particles in air causing harm to life and property.

Major Pollutants: CO, CO₂, SO₂, NO_x, PM₁₀, PM_{2.5}.

Acid Rain: $\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$; $\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_3$.

Control Devices: Settling chamber, Cyclone separator, Bag filter, ESP.

ESP Efficiency: $\eta = 1 - e^{(-Aw/Q)}$.

3. NOISE POLLUTION

Definition: Unwanted sound causing discomfort and health effects.

Unit: Decibel (dB).

Formulas: $L_p = 20 \log(P/P_r)$; $L_w = 10 \log(W/W_0)$.

Effects: Hearing loss, stress, sleep disturbance.

Control: Source control, Path control, Receiver protection.

4. SOLID WASTE MANAGEMENT

Solid waste: Unwanted solid/semi-solid materials from human activities.

Types: Municipal, Industrial, Hazardous.

Disposal: Open dumping, Sanitary landfill, Composting, Incineration, Pyrolysis.

ISWM: Collection → Transportation → Treatment → Disposal.

5. GLOBAL WARMING & CLIMATE CHANGE

Greenhouse Effect: Greenhouse gases trap Earth's heat.

Main gases: CO₂, CH₄, N₂O, CFCs, O₃.

Causes: Fossil fuels, deforestation, industries.

Impacts: Glacier melting, sea rise, droughts, floods.

Mitigation: Renewable energy, afforestation, CO₂ sequestration.

6. OZONE DEPLETION & ACID RAIN

Ozone layer protects Earth from UV radiation.

CFCs destroy ozone molecules.

Acid rain: Rain containing H₂SO₄ and HNO₃ formed from SO₂ and NO_x.

Effects: Plant damage, soil acidity, monument corrosion.

7. FOREST RESOURCES

Direct benefits: Timber, fuel wood, bamboo, medicines.

Indirect benefits: Soil protection, water cycle, climate control, CO₂ absorption.

Deforestation causes: Agriculture, mining, dams, fires.

Effects: Biodiversity loss, soil erosion, global warming.

8. BIODIVERSITY

Definition: Variety of life at genetic, species and ecosystem levels.

Types: Genetic, Species, Ecosystem diversity.

Hotspots in India: Western Ghats, Eastern Himalayas.
Conservation: In-situ and Ex-situ.

9. MINERAL RESOURCES

Minerals: Naturally occurring inorganic crystalline solids.
Uses: Industry, construction, energy generation.
Impacts of mining: Deforestation, water pollution, air pollution.

10. WATER RESOURCES

Hydrological cycle: Evaporation, Transpiration, Precipitation, Infiltration, Runoff.
Water scarcity: Demand exceeds supply.
Sustainable management: Rainwater harvesting, groundwater recharge, drip irrigation.

11. ENVIRONMENTAL LEGISLATION & EIA

Article 48-A: State duty to protect environment.
Article 51-A(g): Citizen duty to protect environment.
Major Acts: Water Act 1974, Air Act 1981, EPA 1986.
EIA: Pre-project environmental impact assessment.

END OF FINAL REVISION NOTES